The Complete Guide to Shavian, Tengwar, and Featural Transliteration

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Part 1: The Shavian Alphabet & ReadLex Specification

The Shavian alphabet is a purely phonemic writing system designed by Kingsley Read to fulfill the bequest of George Bernard Shaw. It replaces traditional non-phonetic English spelling with a 1:1 sound-to-letter mapping.

1. Character Taxonomy

Shavian characters fall into four structural categories based on vertical height and composition:

- **Tall (Voiceless Consonants):** Extend above the x-height.
- **Deep (Voiced Consonants):** Extend below the baseline.
- **Short (Vowels, Liquids, Nasals):** Fit entirely within the x-height.
- **Compound (Vocalic Ligatures):** Combine rhotic (/r/) sounds and vowel clusters into single glyphs.

Consonant Sound Pairs

Tall and Deep consonants form physical pairs where rotating or flipping a Tall (voiceless) letter generates its Deep (voiced) counterpart.

Sound Pair	Tall Glyph (Voiceless)	Deep Glyph (Voiced)	Phonetic Names
/p/ — /b/	𐑐	𐑚	peep — bib
/t/ — /d/	𐑑	𐑛	tot — dead
/k/ — /g/	𐑒	𐑜	kick — gag
/f/ — /v/	𐑓	𐑝	fee — vow
/θ/ — /ð/	𐑔	𐑞	thigh — they
/s/ — /z/	𐑕	𐑟	so — zoo
/ʃ/ — /ʒ/	𐑖	𐑠	sure — measure
/tʃ/ — /dʒ/	𐑗	𐑡	church — judge
/h/ — /w/	𐑣	𐑢	he — woe

Vowels, Liquids, and Compound Ligatures

Category	Characters & Keywords
Liquids & Nasals	𐑤 (/l/ loll), 𐑥 (/m/ mime), 𐑯 (/n/ nun), 𐑙 (/ŋ/ hung), 𐑮 (/r/ roar), 𐑘 (/j/ yea)
Short Vowels	𐑦 (/ɪ/ if), 𐑧 (/ɛ/ egg), 𐑨 (/æ/ ash), 𐑩 (/ə/ ado), 𐑳 (/ʌ/ up), 𐑪 (/ɒ/ on), 𐑫 (/ʊ/ wool)

Category	Characters & Keywords
Long Vowels	𐑰 (/iː/ eat), 𐑱 (/eɪ/ age), 𐑭 (/ɑː/ ah), 𐑷 (/ɔː/ awe), 𐑴 (/oʊ/ oak), 𐑵 (/uː/ ooze)
Diphthongs	𐑲 (/aɪ/ ice), 𐑶 (/ɔɪ/ oil), 𐑬 (/aʊ/ out)
Compound Ligatures	𐑸 (/ɑːr/ are), 𐑹 (/ɔːr/ or), 𐑺 (/ɛər/ air), 𐑻 (/ɜːr/ err), 𐑼 (/ər/ array), 𐑽 (/ɪər/ ear), 𐑾 (/iə/ Ian), 𐑿 (/juː/ yew)

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2. The 12 ReadLex Orthographic Rules

1. **Strict Sound-to-Letter Phonetics:** Write strictly by spoken sound, ignoring silent Latin letters (e.g., *write* = `𐑮𐑲𐑑`).
2. **Phonetic Voicing Assimilation in Suffixes:** Suffixes match spoken voicing (e.g., *cats* = `𐑒𐑨𐑑𐑕` vs. *dogs* = `𐑛𐑪𐑜𐑟`; *walked* = `𐑢𐑷𐑒𐑑` vs. *played* = `𐑐𐑤𐑱𐑛`).
3. **The Big Five Abbreviations:** Five high-frequency function words are abbreviated as single shorthand letters: *the* = **𐑞**, *of* = **𐑝**, *and* = **𐑯**, *to* = **𐑑**, *for* = **𐑓**.
4. **Monosyllabic Invariance:** Single-syllable words maintain their fully stressed spelling regardless of context (e.g., *but* is always `𐑚𐑳𐑑`).
5. **Monosyllabic Weak Vowel Exception:** Indefinite articles are explicitly written weak using schwa (`𐑩`): *a* = `𐑩`, *an* = `𐑩𐑯`.
6. **Stress Distinction in Polysyllabic Words:** Vowel reduction reflects stress shifts (e.g., noun *convict* = `𐑒𐑪𐑯𐑝𐑦𐑒𐑑` vs. verb *convict* = `𐑒𐑩𐑯𐑝𐑦𐑒𐑑`).
7. **No Letter Gemination:** Consonants are never doubled unless articulated twice across syllable boundaries (e.g., *annoy* = `𐑩𐑯𐑶` vs. *unnamed* = `𐑳𐑯𐑯𐑱𐑥𐑛`).
8. **Syllabic Consonant Formatting:** Insert a schwa (`𐑩`) before terminal liquids or nasals (e.g., *little* = `𐑤𐑦𐑑𐑩𐑤`, *driven* = `𐑛𐑮𐑦𐑝𐑩𐑯`).
9. **Verbal Contraction Formatting:** Negative contractions insert `𐑩` before `𐑯𐑑` and omit apostrophes (e.g., *didn't* = `𐑛𐑦𐑛𐑩𐑯𐑑`).
10. **Proper Noun Marking:** Capital letters are omitted entirely; proper nouns use a middle Namer Dot (`·`) immediately preceding the word (e.g., *London* = `·𐑤𐑳𐑯𐑛𐑩𐑯`).
11. **Unstressed Terminal Vowels (*happY* Rule):** Words ending in unstressed *-y* or *-ie* use `𐑦` (e.g., *happy* = `𐑣𐑨𐑐𐑦`). Tense, fully stressed final vowels use `𐑰` (e.g., *trustee* = `𐑑𐑮𐑳𐑕𐑑𐑰`).
12. **Cross-Dialectal "Rhotic RP" Standard:** Standard spelling uses a broad rhotic Received Pronunciation base so rhotic and non-rhotic speakers share a single format.

Part 2: Tolkien's Tengwar & Alcarin Tengwar Standard

Tengwar is a featural abugida created by J.R.R. Tolkien. Primary consonants (*tengwar*) represent structural articulation features, while vowels (*tehtar*) sit above or below consonant bases as diacritics.

1. The Fëanorian Consonant Grid (Appendix E Layout)

In Tolkien's standard layout, primary consonants are organized into **4 Series** (Columns: Place of Articulation) and **6 Grades** (Rows: Manner of Articulation and Voicing):

- **Series I (*Témar 1*):** Dental / Alveolar (/t/, /d/, /θ/, /ð/)
- **Series II (*Témar 2*):** Labial / Bilabial (/p/, /b/, /f/, /v/)
- **Series III (*Témar 3*):** Palatal / Affricate (/ʧ/, /ʤ/, /ʃ/, /ʒ/)
- **Series IV (*Témar 4*):** Velar (/k/, /g/, /ŋ/)

Grade / Manner	Series I (Dental)	Series II (Labial)	Series III (Palatal)	Series IV (Velar)
Grade 1: Voiceless Stops *(Stem Down, Single Loop)*	 `U+E000` *Tinco* (/t/)	 `U+E001` *Parma* (/p/)	 `U+E002` *Calma* (/ʧ/)	 `U+E003` *Quesse* (/k/)
Grade 2: Voiced Stops *(Stem Down, Double Loop)*	 `U+E004` *Ando* (/d/)	 `U+E005` *Umbar* (/b/)	 `U+E006` *Anga* (/ʤ/)	 `U+E007` *Ungwe* (/g/)
Grade 3: Voiceless Fricatives *(Stem Up, Single Loop)*	 `U+E008` *Thúle* (/θ/)	 `U+E009` *Formen* (/f/)	 `U+E00A` *Harma* (/ʃ/)	 `U+E00B` *Hwesta* (/hw/)
Grade 4: Voiced Fricatives *(Stem Up, Double Loop)*	 `U+E00C` *Anta* (/ð/)	 `U+E00D` *Ampa* (/v/)	 `U+E00E` *Anca* (/ʒ/)	 `U+E00F` *Unque* (/gh/)
Grade 5: Nasals *(Short Stem, Double Loop)*	 `U+E010` *Númen* (/n/)	 `U+E011` *Malta* (/m/)	 `U+E012` *Noldo* (/ŋ/)	 `U+E013` *Nwalme* (/ŋ/)
Grade 6: Glides / Semi-vowels *(Short Stem, Single Loop)*	 `U+E014` *Óre* (not used by this mode)	 `U+E015` *Vala* (/w/)	 `U+E016` *Anna* (/j/)	 `U+E017` *Vilya*

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2. Additional & Irregular Tengwar (Alcarin Standard)

Beyond the standard 24-tengwa grid, specialized base characters handle liquids, sibilants, glottal sounds, and silent carriers:

Tengwa Glyph	Code Point	Tengwa Name	Structural Description / Function
	`U+E020`	*Rómen*	/r/ in every position (R-01)
	`U+E022`	*Lamba*	Upward hooked liquid character (/l/)
	`U+E024`	*Silme*	Upright S-curve sibilant (/s/)
	`U+E025`	*Silme Nuquerna*	Inverted S-curve (used when carrying top tehta)
	`U+E026`	*Esse / Aze*	Z-curve / inverted double loop (/z/)
	`U+E027`	*Esse Nuquerna*	Inverted Z-curve (used when carrying top tehta)
	`U+E028`	*Hyarmen*	Downward stem with left hook (/h/)
	`U+E02C`	***Ára* (Long Carrier)**	Descending vertical stem (dotless 'j', drops below baseline) carrying long orphan vowels
	`U+E02D`	*Halla* (unused by engine)	Ascending vertical stem (starts above x-height); not a carrier
	`U+E02E`	***Telco* (Short Carrier)**	Neutral x-height vertical stem (dotless 'i') carrying short orphan vowels

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3. Vowel Tehtar & Diacritic Marks

Vowel marks (*tehtar*) sit above the *following* consonant and are read before it. A long vowel is the same mark as its short partner (KIT/FLEECE, TRAP/PALM, LOT/THOUGHT, FOOT/GOOSE) but written on the long carrier `{aara}` instead of on a consonant. STRUT and the schwa have marks of their own.

Mark	Code point	Name	Used for	Placement
	`U+E040`	*a-tehta*	TRAP; PALM; the first half of PRICE and MOUTH	three dots in a triangle above
	`U+E044`	*i-tehta*	KIT, FLEECE	single dot above
	`U+E046`	*e-tehta*	DRESS, FACE	single acute stroke above
	`U+E04A`	*o-tehta*	LOT, THOUGHT, CHOICE, GOAT	right-facing curl above
	`U+E04C`	*u-tehta*	FOOT, GOOSE	left-facing curl above
	`U+E053`	*schwa-tehta*	the schwa /ə/	breve above
	`U+E054`	*grave*	STRUT	grave stroke above
	`U+E050`	*nasal bar*	nasal plus stop (ʅ1, ʃ1 …), ℓძ, ℓϼ	bar above the stop
	`U+E051`	*doubler*	two identical consonants	bar below the consonant
	`U+E045`	*under-dot*	only inside the *and* shorthand	dot below

The double acute (`U+E048`), a doubled left curl and the three inverted dots (`U+E056`) are not used. When a mark lands on Silmë or Essë, the nuquerna form is used instead (S-02). Mark order on a base is vowel, then nasal bar, then doubler (K-01).

Part 3: The Shavian-to-Tengwar Transliteration Bridge

By using Shavian as an Intermediate Representation (IR), transliteration bypasses non-phonetic English orthography entirely. Each Shavian word becomes a list of cells (one base glyph plus its marks), which an encoding then turns into code points.

The rules themselves live in `spec/mode.md`; this part is a reader's summary. The tables below were generated from the engine, so they show what the converter actually writes. If they disagree with the spec, the spec wins.

1. Consonants

Each consonant letter has one fixed glyph (C-01). The Rules column lists the spec rules that fired.

Sound	Shavian	Tengwar	Glyph names	Code points	Rules
/p/	𐑐		`{parma}`	`U+E001`	C-01
/b/	𐑚		`{umbar}`	`U+E005`	C-01
/t/	𐑑		`{tinco}`	`U+E000`	C-01
/d/	𐑛		`{ando}`	`U+E004`	C-01
/k/	𐑒		`{quesse}`	`U+E003`	C-01
/g/	𐑜		`{ungwe}`	`U+E007`	C-01
/tʃ/	𐑗		`{calma}`	`U+E002`	C-01
/dʒ/	𐑡		`{anga}`	`U+E006`	C-01
/f/	𐑓		`{formen}`	`U+E009`	C-01
/v/	𐑝		`{ampa}`	`U+E00D`	C-01
/θ/	𐑔		`{thuule}`	`U+E008`	C-01
/ð/	𐑞		`{anto}`	`U+E00C`	C-01
/ʃ/	𐑖		`{harma}`	`U+E00A`	C-01
/ʒ/	𐑠		`{anca}`	`U+E00E`	C-01
/s/	𐑕		`{silme}`	`U+E024`	C-01
/z/	𐑟		`{esse}`	`U+E026`	C-01
/m/	𐑥		`{malta}`	`U+E011`	C-01
/n/	𐑯		`{nuumen}`	`U+E010`	C-01
/ŋ/	𐑙		`{nwalme}`	`U+E013`	C-01
/l/	𐑤		`{lambe}`	`U+E022`	C-01

Sound	Shavian	Tengwar	Glyph names	Code points	Rules
/r/	𐑮		`{roomen}`	`U+E020`	C-01
/w/	𐑢		`{vala}`	`U+E015`	C-01
/j/	𐑘		`{anna}`	`U+E016`	C-01
/h/	𐑣		`{hyarmen}`	`U+E028`	C-01

Special cases: √ is the one glyph `{hwesta}` (C-02). ʅ1, ʅɻ, ʃ1, ʃɭ, ℓƈ and ℓɔ are the stop plus a nasal bar (N-01, N-02). Two identical consonants take the doubling bar below (G-01). Silmë and Essë flip to their nuquerna forms when a mark lands on them (S-02).

2. Vowels

A short vowel is a mark on the *next* consonant, read before it. A short vowel with no consonant after it sits on the short carrier `{telco}`. A long vowel always sits on the long carrier `{aara}` (V-21..V-24). The "alone" forms below show that orphan case.

Sound	Shavian	Tengwar	Glyph names	Code points	Rules
/ɪ/	𐑦		`{telco}[dot-above]`	`U+E02E U+E044`	V-11, V-02
/ɛ/	𐑧		`{telco}[acute]`	`U+E02E U+E046`	V-12, V-02
/æ/	𐑨		`{telco}[triple-dot-above]`	`U+E02E U+E040`	V-13, V-02
/ə/	𐑩		`{telco}[breve]`	`U+E02E U+E053`	V-14, V-02
/ɒ/	𐑪		`{telco}[right-curl]`	`U+E02E U+E04A`	V-15, V-02
/ʌ/	𐑳		`{telco}[grave]`	`U+E02E U+E054`	V-16, V-02
/ʊ/	𐑫		`{telco}[left-curl]`	`U+E02E U+E04C`	V-17, V-02
/iː/	𐑰		`{aara}[dot-above]`	`U+E02C U+E044`	V-21, V-03
/ɑː/	𐑭		`{aara}[triple-dot-above]`	`U+E02C U+E040`	V-22, V-03
/ɔː/	𐑷		`{aara}[right-curl]`	`U+E02C U+E04A`	V-23, V-03
/uː/	𐑵		`{aara}[left-curl]`	`U+E02C U+E04C`	V-24, V-03

3. Diphthongs and R-vowels

The glide is written last and the vowel mark first. R-vowels are not separate glyphs: they expand to a vowel plus 𐑮 (rómen). 𐑽 is 𐑦𐑩 and 𐑿 is 𐑘𐑵.

Sound	Shavian	Tengwar	Glyph names	Code points	Rules
/aɪ/	𐑲		`{anna}[triple-dot-above]`	`U+E016 U+E040`	D-01
/aʊ/	𐑬		`{vala}[triple-dot-above]`	`U+E015 U+E040`	D-02
/eɪ/	𐑱		`{anna}[acute]`	`U+E016 U+E046`	D-03
/oʊ/	𐑴		`{vala}[right-curl]`	`U+E015 U+E04A`	D-04
/ɔɪ/	𐑶		`{anna}[right-curl]`	`U+E016 U+E04A`	D-05

Sound	Shavian	Tengwar	Glyph names	Code points	Rules
/juː/	𐑿		`{anna}{aara}[left-curl]`	`U+E016 U+E02C U+E04C`	L-07, C-01, V-24, V-03
/ɪə/	𐑾		`{telco}[dot-above]{telco}[breve]`	`U+E02E U+E044 U+E02E U+E053`	L-08, V-11, V-02, V-14
/ɑːr/	𐑸		`{aara}[triple-dot-above]{roomen}`	`U+E02C U+E040 U+E020`	L-01, V-22, V-03, R-01
/ɔːr/	𐑹		`{aara}[right-curl]{roomen}`	`U+E02C U+E04A U+E020`	L-02, V-23, V-03, R-01
/ɛər/	𐑺		`{roomen}[acute]`	`U+E020 U+E046`	L-03, R-01, V-12, V-01
/ɪər/	𐑽		`{roomen}[dot-above]`	`U+E020 U+E044`	L-04, R-01, V-11, V-01
/ɜːr/	𐑻		`{roomen}[grave]`	`U+E020 U+E054`	L-05, R-01, V-16, V-01
/ər/	𐑼		`{roomen}[breve]`	`U+E020 U+E053`	L-06, R-01, V-14, V-01

4. The Big Five: Shorthand Logograms

Shavian writes five very common words as a single letter. *the*, *of* and *and* have Tengwar shorthands. *to* and *for* do not: they are written as their weak forms /tə/ and /fər/, which keeps them distinct from *too/two* and *four*. This replaces an older version of this guide that gave *to* and *for* as the bare letters Tinco and Formen (see `DECISIONS.md`, D-01).

Target word	Shavian	Tengwar	Glyph names	Code points	Rules
the	𐑞		`{the-shorthand}`	`U+E01C`	W-01
of	𐑝		`{of-shorthand}`	`U+E01D`	W-02
and	𐑯		`{and-shorthand}`	`U+E004 U+E050 U+E045`	W-03
to	𐑑		`{tinco}{telco}[breve]`	`U+E000 U+E02E U+E053`	W-04, L-09, C-01, V-14, V-02
for	𐑓		`{formen}{roomen}[breve]`	`U+E009 U+E020 U+E053`	W-04, L-10, C-01, L-06, R-01, V-14, V-01

5. Punctuation

Input	Glyph names	Code point	Rule
`,`	`{pusta}`	`U+E060`	U-01
`.`	`{double-pusta}`	`U+E061`	U-01
`;`	`{double-section-mark}`	`U+E069`	U-02

Input	Glyph names	Code point	Rule
`:`	`{triple-pusta}`	`U+E062`	U-03
`-`	`{section-mark}`	`U+E068`	U-04
`!`	`{exclamation-mark}`	`U+E065`	U-05
`?`	`{question-mark}`	`U+E066`	U-05
`(`	`{parenthesis-mark}`	`U+E067`	U-05
`)`	`{parenthesis-mark}`	`U+E067`	U-05
`“`	`{left-quotation-mark}`	`U+E06A`	U-05
`”`	`{right-quotation-mark}`	`U+E06B`	U-05
`"` (ASCII)	alternates the two quotation marks		U-05